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How to Become a Statistician in 2024

Curious about how to become a statistician? Find out what a statistician does, what you need to get started, and what you can expect from this career.

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Joleen Bothma

Data Science Consultant & Writer at Statistically Relevant

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We live in a time where more data is available than ever, and we continue to collect more and more data every minute. As such, the demand for experts like statisticians to analyze this data and develop data-driven solutions to business and research problems is at its highest.



So, how do you become a statistician? First, you need to know what a statistician does, what you need to start a career as a statistician, and what you can expect from this career in terms of salary and job outlook. This article will answer all these questions and more.

You can check out other articles in our series to learn [how to become a quantitative analyst](#), [a data scientist](#), [a data engineer](#), or [a data analyst](#).

What Is a Statistician?

Statisticians are problem-solvers. They collect data and use statistical modeling and data manipulation techniques in specialized software to develop solutions and forecast the future. Their work involves not just statistical analyses and data but also communication, collaboration, and business understanding.

Statisticians can work in applied or theoretical statistics. Applied statistics involves collecting and analyzing data to understand/explain the real world or predict future events. Applied statisticians can be employed in nearly any industry with data and a problem to solve.

On the other hand, theoretical statistics involves the mathematical justification of the models and methods applied by statisticians. Theoretical statisticians are primarily involved in research and academia.

As more data is collected and becomes available to organizations in many different industries, the demand for statisticians also increases. Organizations need statisticians to analyze and interpret what all that data means and drive their decision-making. Even research institutions rely on statisticians to analyze their data and back up their research claims and hypotheses.

To learn more about statistics, check out our [Introduction to Statistics](#) course. For applications using the R programming language, you can also check out the [Introduction to Statistics in R](#) course.

What Do Statisticians Do?

Applied statisticians, the most common and sought-after statistical profession, involves the application of statistical methods to problems to develop data-driven solutions. Here, you'll get a better idea of what is involved in the job of an (applied)

statistician, from the job description to their roles and responsibilities in an organization.

Statistician job description

As a statistician, you will:

- **Analyze and interpret data** to help businesses, governments, and other institutions to solve problems and make decisions
- **Define the problem to be solved** (in some cases), identify the data needed to solve it, and choose the most appropriate statistical method.
- **Use probability, calculus, statistical modeling, and programming** in your work.
- **Analyze data to find trends, patterns, and relationships** using your technical expertise and business understanding.
- **Communicate and present your findings** to stakeholders.
- **Explain your findings and the process you followed to reach your results.** You must be able to explain or show how it benefits or adds value to the organization. This often involves both technical and non-technical audiences.

Statistician roles and responsibilities

Statisticians play a crucial role in businesses, governments, and other institutions. As such, they are often highly-regarded, senior members of staff. This is because a statistician's extensive requirements and job description usually mean they are highly qualified and compensated well for their technical expertise (more on salary expectations later in the article).

Statisticians may work alone, within a team, or even run a team entirely. Of course, this depends on the organization they are working for.

Statisticians in a research role typically work as the sole statisticians on a project but need to collaborate closely with people in other fields and specialties.

What Do You Need to Become a Statistician?

You will require at least a formal qualification to get a job as a statistician. Certifications are available, but they are usually optional and may only be required in very specialized areas of work.

Qualifications

Most statisticians require at least a university degree in STEM (Science, Technology, Engineering, and Math). However, some even require a postgraduate degree, such as a Master's or PhD.

The most desirable qualification to obtain is a statistics degree (or above). This degree will naturally provide you with the most relevant knowledge and skills relating to the job of a statistician.

However, it is also possible to get a job as a statistician with a degree in another area, such as in Physics or Computer Science. Ultimately, the job of a statistician requires a lot of experience. If you can demonstrate that you have several years of experience working with and analyzing data, it is likely that your degree will not matter as much.

There are many additional certifications that you can add to your resume to stand out to potential employers and demonstrate proficiency in specific areas or with specialized software. Some examples of these are:

- **American Statistical Association:** PSTAT and GTSTAT accreditation routes
- SAS and SPSS
- Python and R certifications. Prepare with the **Statistics Fundamentals with Python** and **R** DataCamp skill tracks.

Skills

As a statistician, you will require both technical and soft skills to do your job well.

Statistician technical skills

Some of the technical skills you may require are:

- Applied and theoretical statistical methods.
- Mathematics, like linear algebra and calculus.

- Data mining and transformation to uncover insights in the data and prepare the data for statistical modeling.
- Data visualization—This includes data visualization techniques and knowledge of some of the various specialized software commonly used to visualize data (e.g., Power BI, Tableau, Matplotlib, Ggplot, etc.).
- Computer science specialized skills: machine learning, **artificial intelligence**, programming, and computing infrastructure.

Specifically, most statisticians must know one or more programming languages and/or specialized statistical software. Examples of these are:

- **Programming languages:** R, Python, Java, Perl, Julia.
- **Specialized software:** SAS, SPSS, Matlab.
- **Database software:** SQL Server, PostgreSQL, and NoSQL.

The R programming language is one of the most popular languages amongst statisticians, so you will likely need to know this language to land a job as a statistician. However, depending on the industry, specialized software like SAS and SPSS are a requirement (this is common in certain research institutions and governments).

Database software is essential for statisticians, particularly in organizations that collect and store large amounts of data. As a statistician, you will need to determine what data you will need to solve a business problem. You may also need to extract data from databases or warehouses, depending on whether the organization has dedicated teams for that or not.

Statistician soft skills

Some of the soft skills you may require as a statistician are:

- **Communication.** Empathy, understanding, and a willingness to listen are essential elements of effective communication. Additionally, you need to be able to effectively explain the results of your work to a non-technical audience and demonstrate how it can add value to their job and to the organization.

- **Teamwork.** Working as a team and collaborating with people from different parts of an organization are crucial to doing your job well as a statistician. Data analyses and statistical modeling cannot happen in a vacuum; you will need to rely on the expertise and skills of other individuals to find the best solutions.
- **Curiosity.** An innate sense of curiosity about why things are the way they are will drive your analyses and search for solutions. Constantly going deeper and deeper into the data will allow you to expand your perspective and not get bogged down by pre-existing assumptions about what 'should' be happening. Instead, you can discover new and unique insights.

Statistician Career Paths

Statisticians typically choose a specialty in either a field or a function, which dictates the types of tools and methods they commonly use to solve the highly specific problems they encounter in that field.

For example, statisticians specializing in economics are called econometricians, and statisticians specializing in biology are called biostatisticians.

Statisticians are employed in a variety of different fields and industries. However, these are the primary industries that maintain a high demand for statisticians:

- **Research and academia.** This is one of the most varied roles a statistician can take on. You might work on a single project over a few years or jump from project to project, providing statistical analyses that back up research hypotheses. Your role in academia also incorporates teaching and supervising students.
- **Government.** Based on data collected by cities and countries, your role in this industry may involve analyses on crime, traffic, weather, housing, and even the best sectors for the government to increase expenditure in. The primary role of a statistician in this area is to offer insights that governments can use to better inform their spending, decision-making, and legislature.
- **Healthcare.** In this industry, you can become a biostatistician, which may involve many research-based statistical analyses.
- **Sports.** The sports industry has become heavily dependent on the expertise of a statistician for everything from recruiting new team members to optimizing their

gameplay to increase the chance of winning.

In addition, a few career paths are similar to that of the statistician, such as the data scientist and data analyst. However, while these careers require similar qualifications and skills, there is a greater emphasis on programming ability, extensive knowledge of computer science concepts, and the application of artificial intelligence to solving business problems.

Statistician Salary and Job Outlook

Job growth is overwhelmingly positive for statisticians and has been this way for many years. According to the [US Bureau of Labor Statistics](#), "Overall employment of mathematicians and statisticians is projected to grow 31 percent from 2021 to 2031, much faster than the average for all occupations."

They also state that the 2023 median pay for statisticians is around \$104,860 per



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However, obtaining the wage values mentioned here requires at least 2-4 years of experience as a statistician, often more. So be prepared to climb the ranks over several years to reach this higher pay level.

Conclusion

With a high salary expectation and positive job outlook, a career as a statistician is a popular and worthwhile choice. Moreover, now that you have a good understanding of what a career as a statistician is all about, you can gain the knowledge and skills you need to become a statistician using the DataCamp career track [Statistician with R](#).

Your Path to Mastering R

Start from scratch and build core R skills for data science.